# FusRock® FDM Printing Material
# Technical Data Sheet

**Data / Revised: 08.2023** **Version No: 5.0**

## FusForce™ PC-FR

阻燃 PC/ABS 3D 打印材料

**Flame Retardant PC/ABS alloy 3D Printing Filament**

## 产品亮点

## Product highlights

- **无卤阻燃**

  FusForce™ PC/ABS-FR 是一款优秀的防火材料，以 PC/ABS 基材，通过添加无卤阻燃剂，使材料具有优秀的自熄性能，阻燃等级可达 UL94 V-0 级（2mm）。相比于添加卤素阻燃剂的材料，FusForce™ PC/ABS-FR 燃烧后不会释放大量烟雾和有毒气体，更加安全环保，同时仍然保留了 PC/ABS 优秀的机械性能和耐热性。

- **Halogen-free flame retardants**

  **FusForce™ PC/ABS-FR is an excellent flame-resistant material which is based PC/ABS. By adding non-halogen flame retardant, PC/ABS has excellent self-extinguishing performance, and the flame-retardant grade can reach UL94 V-0 level (2mm). Compared with materials with halogen flame retardants, FusForce™ PC/ABS-FR will not release much smoke and toxic gases after burning. Therefore, FusForce™ PC/ABS-FR is safer and more environmentally friendly, while still maintaining the excellent mechanical properties and heat resistance of PC/ABS.**

## 产品详情

## Product details

颜色 Color：自然色 Natural

线径 Diameter：1.75mm

净重 Net Weight：500g，1kg，2.5kg

## 物性表（v1.0）

## Material Properties

| 测试项目<br>Property | 测试方法<br>Test Method | 典型值<br>Typical Value |
|---|---|---|
| 密度<br>Density | ISO 1183 | 1.19 g/cm³ |
| 玻璃化转变温度<br>Glass transition temperature | ISO 11357 | 105°C |
| 熔融指数<br>Melt index | 250°C，2.16kg | 25 g/10min |
| 热变形温度（X-Y）<br>Heat deflection temperature (X-Y) | ISO 75：Method A<br>ISO 75：Method B | 83°C (1.8MPa)<br>88°C (0.45MPa) |
| 拉伸屈服强度（X-Y）<br>Tensile yield strength (X-Y) | ISO 527 | 52.51 ± 0.28 MPa |
| 屈服点伸长率（X-Y）<br>Elongation at Yield (X-Y) | | 3.38 ± 0.02 % |
| 杨氏模量（X-Y）<br>Young's modulus (X-Y) | | 2588.73 ± 64.81 MPa |
| 拉伸断裂强度（X-Y）<br>Tensile breaking strength (X-Y) | | 48.99 ± 1.20 MPa |
| 断裂伸长率（X-Y）<br>Elongation at break (X-Y) | | 5.55 ± 0.99 % |
| 拉伸断裂强度（Z）<br>Tensile breaking strength (Z) | ISO 527 | 34.26 ± 0.84 MPa |
| 杨氏模量（Z）<br>Young's modulus (Z) | | 2302.81 ± 46.92 MPa |
| 断裂伸长率（Z）<br>Elongation at break (Z) | | 1.88 ± 0.11 % |
| 弯曲强度（X-Y）<br>Bending strength (Z) | ISO 178 | 85.95 ± 0.83 MPa |
| 弯曲模量（X-Y）<br>Bending modulus (X-Y) | | 2504.55 ± 22.88 MPa |

| | | |
|---|---|---|
| 缺口冲击强度（X-Y）<br>Charpy impact strength (X-Y) | ISO 179 | 8.39 ± 0.46 kJ/㎡ |
| UL 阻燃等级<br>UL Flame-retardant Grade | | |
| 2mm 厚度<br>2mm thickness | UL94 | V-0 |
| 2.5mm 厚度<br>2.5mm thickness | | 5VB |

试样打印参数：喷嘴大小 0.4mm，喷嘴温度 250°C，底板加热 110°C，腔体温度 55℃，打印速度 45mm/s，填充率 100%，填充角度±45°

退火条件：90℃，退火 4 小时

**Specimens printed under the following conditions: Nozzle size 0.4mm, Nozzle temp 250°C, Bed temp 110°C, Chamber temp 55°C, Printing speed 45mm/s, Infill 100%, Infill angle ±45°.**

## 建议打印参数

## Recommended printing conditions

| | |
|---|---|
| 喷头温度<br>Nozzle temperature | 240-270°C |
| 建议喷嘴大小<br>Recommended Nozzle diameter | ⩾0.2mm |
| 建议底板材质<br>Recommended build surface treatment | 玻璃、PEI 膜或 PC 膜<br>Glass、PEI Film or PC Film |
| 底板温度<br>Build plate temperature | 100-110°C |
| 腔体温度<br>Chamber temperature | 40-70°C |
| Raft 间距<br>Raft separation distance | 0.18-0.2 mm |
| 冷却风扇<br>Cooling fan speed | 0-20% |
| 打印速度<br>Printing speed | 30-300 mm/s |
| 回抽距离 | 1-3 mm |

| | |
|---|---|
| Retraction distance | |
| 回抽速度<br>Retraction speed | 1800-3600 mm/min |

其他建议：

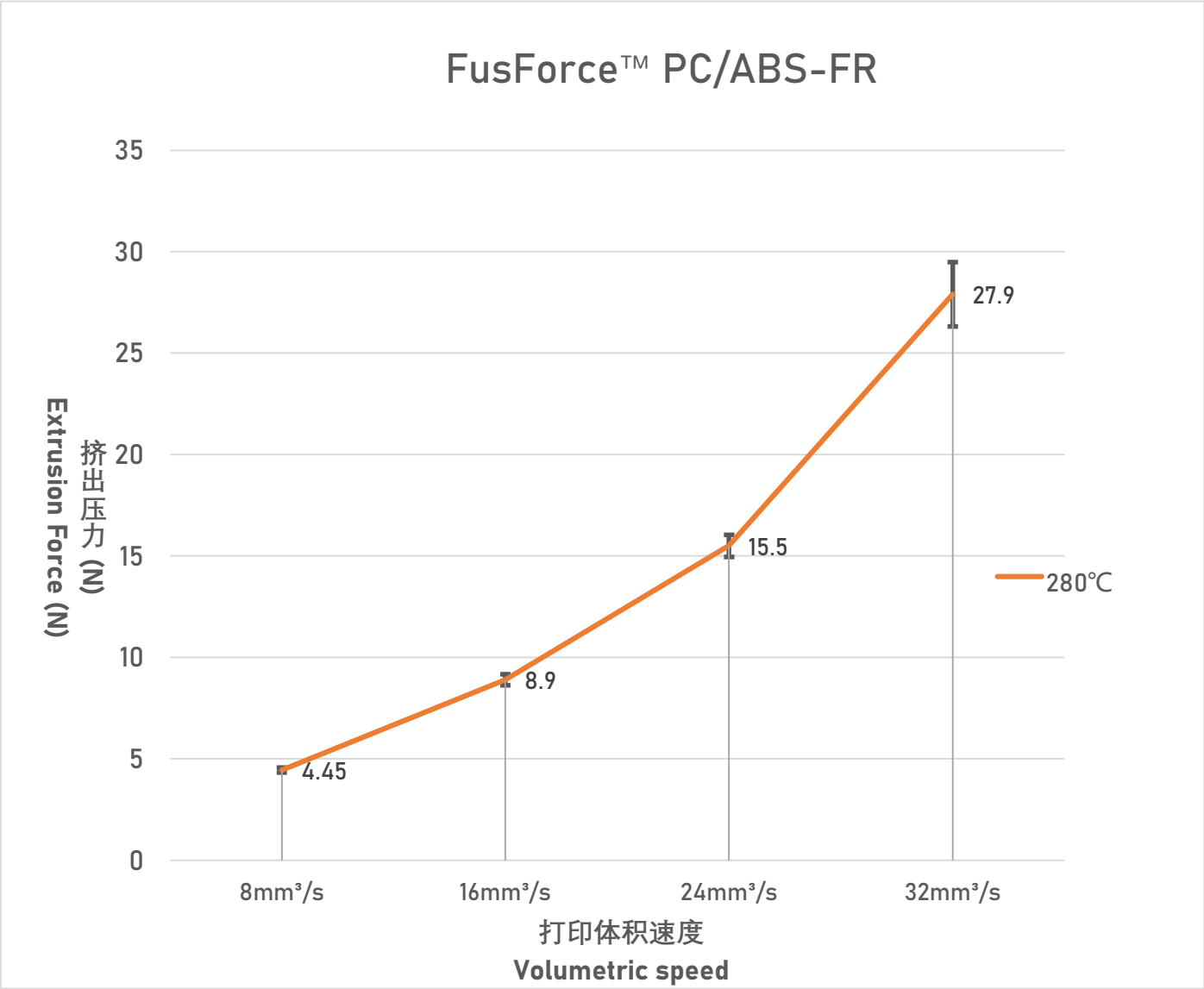
1. PC/ABS-FR 打印温度较高，对水分较敏感，在打印过程中将线材放入干燥盒内，相对湿度控制在 15%以下，可有效减少拉丝、表面粗糙等现象。如果耗材已吸潮，可以使用烘箱在 75℃进行 4-6 小时的烘干处理。
2. 建议在打印 PC/ABS-FR 耗材时将 3D 打印机放置在通风环境中。

**Additional suggestions：**

1. **PC/ABS-FR filament is sensitive to moisture and can easily have stringing during printing. To reduce the stringing and surface roughness effectively, keep your filament in a dry box and control relative humidity to below 15%. If the filament has absorbed moisture already, dry the filament in an oven at 75°C for 4-6 hours.**
2. **It is recommended to place the printer in a well-ventilated area during printing PC/ABS-FR filament.**

## 挤出压力与打印流量速度测试

## Extrusion Force vs Print Volumetric Speed Test




测试参数：12mm 长度铜制加热块，BMG 挤出机，Phaetus 硬化钢喷头，喷嘴大小 0.4mm，层高 0.2mm。

**Test parameters: 12mm length brass heat block, BMG extruder, Phaetus Hardened Steel Nozzle, Nozzle size 0.4mm, Layer Height 0.2mm.**